HYDROMETALLURGY

METALS EXTRACTION. SUSTAINABLE FUTURE.

Hydrometallurgy uses aqueous solutions and chemical processes to extract and recover metals from ores, concentrates, and secondary resources in an efficient and environmentally responsible way.

- Lower Energy Use
- High Metal Recovery
- Resource Efficient
- Cleaner Process

INDUSTRIAL PROCESS FLOW (TYPICAL COPPER SX–EW CIRCUIT)

1 ORE RECEIVING & HANDLING

- Ore unloading, storage & blending
- Sampling & characterization

2 PRETREATMENT

- Crushing
- Grinding
- Agglomeration
- (Option) Roasting / Oxidative pretreatment

3 LEACHING

Metals are dissolved into solution using suitable lixiviants.

- Heap leaching (H_2SO_4)
- Tank / Agitated leaching
- Pressure leaching
- Alkaline / Chloride leaching

4 SOLID–LIQUID SEPARATION

Separation of pregnant leach solution (PLS) from residual solids.

- Thickening
- Filtration
- Washing of residue

5 SOLIUTION PURIFICATION

Removal of impurities and separation of valuable metals.

- Solvent Extraction (SX)
- Ion Exchange (IX)
- Adsorption
- Precipitation / Cementation (if needed)

6 METAL RECOVERY

Recovery of metals from purified solutions.

- Electrowinning (EW)
- Precipitation / Cementation
- Reduction (e.g., H_2)

7 PRODUCT REFINING

Purification of recovered metals to achieve required grade.

- Melting
- Electrorefining
- Chemical refining

8 WASTE & WATER MANAGEMENT

Treatment and recycling for sustainable operations.

- Effluent treatment
- Water recycling
- Residue management

KEY HYDROMETALLURGICAL METHODS

ACID LEACHING

Uses acids (e.g., H_3SO_4, HCl) to dissolve metals from ores. Common for Cu, Ni, Zn, Co, U.

ALKALINE LEACHING

Uses bases (e.g., NaOH, NH_3) for metals like Al (Bayer process), Au, and some rare earths.

SOLVENT EXTRACTION (SX)

Selective transfer of metals from aqueous phase to organic phase using extractants.

ION EXCHANGE (IX)

Ions are adsorbed on a resin and later eluted to recover metals in concentrated form.

PRECIPITATION / CEMENTATION

Metals are precipitated or displaced by a more reactive metal (e.g., Zn cementation).

ELECTROWINNING (EW)

Metals are deposited on a cathode from solution using electric current.

MAJOR APPLICATIONS

Copper, Nickel, Cobalt extraction

Uranium and Rare Earth metals recovery

Gold & Silver recovery

E-Waste & Spent Battery recycling

Zinc, Manganese, Lithium recovery

Industrial effluent treatment & resource recovery

KEY ADVANTAGES

- Lower energy consumption compared to pyrometallurgy
- High metal recovery & selectivity
- Suitable for low-grade ores & complex materials
- Lower greenhouse gas emissions
- Enables recycling and circular economy
- Safer working conditions

TYPICAL INDUSTRIES

Mining & Mineral Processing

Non-Ferrous Metal Industry

Battery & E-Waste Recycling

Chemical & Fertilizer Industry

Water & Wastewater Treatment

Renewable & Clean Tech Supply Chain

ENVIRONMENTAL IMPACT ANALYSIS

POSITIVE IMPACTS

- Lower greenhouse gas emissions than pyrometallurgical routes
- Lower energy requirements
- Less air pollution (SO_x, NO_x, dust) when well managed
- Water can be recycled and reused in closed-loop systems
- Enables recovery of metals from secondary resources and wastes
- Supports circular economy and resource efficiency

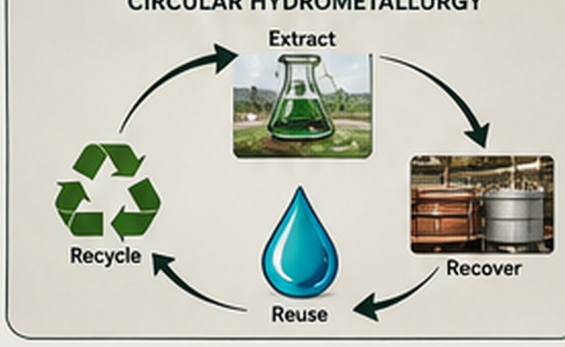
POTENTIAL ENVIRONMENTAL CONCERNS

- High water consumption if recycling is not implemented
- Use of acids, bases, and organic solvents can pose environmental risks
- Generation of acidic or metal-bearing effluents
- Improper disposal can lead to soil and groundwater contamination
- Solid residues and tailings require safe management
- Energy use in pumping, heating, and electrowinning

SUSTAINABLE SOLUTIONS

- Closed-loop water systems & zero liquid discharge (ZLD)
- Green lixiviants & less haaardous reagents
- Advanced solvent extraction with low-toxicity extractants
- Membrane separation & adsorption technologies
- Process optimization, automation & digital monitoring
- Safe residue stabilization & beneficial reuse

CIRCULAR HYDROMETALLURGY



THE FUTURE

Hydrometallurgy is a key enabler of the circular economy and critical for a sustainable, resource-secure world.

CLEANER PROCESS.
GREENER PLANET.
STRONGER TOMORROW.